

GIS bio-waste assessment and suitability analysis for biogas power plant: A case study of Anambra state of Nigeria



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ABSTRACT

Environmental concerns have increased over environmental hazards associated with fossil energy source, transition to renewable energy sources have been emphasized, this however requires relevant data for informed decision making. Critical analytical methods in situating bioenergy plants for bio-energy production are advantageous from environmental, economic and sustainable perspectives. A study therefore was undertaken to provide relevant data on biogas renewable energy source using Anambra State of Nigeria as a case study. The study was carried out using the power of Geographical Information System (GIS) technology. Several GIS thematic layers were obtained considering important factors in bioenergy plant location analysis. GIS-based thematic maps used for the production of the suitability map include political boundary map layer, road network layer, Land Use and Land Cover (LULC) layer, Digital Elevation Map (DEM) layer, river layer, slope layer, electric transmission line network layer, biomass potential density layer etc. Considering the fact that bioenergy plant is dependent on heat energy source which is also abundant in the tropics, hill-shade layer was integrated into the GIS location suitability analysis. Various economic and socio-environmental factors were considered and utilized in the development of economic, land suitability and bio-resource spatial density maps. These thematic maps were overlaid to obtain the final suitability map index. The most suitable sites identified were located in Njikoka, Onitsha North and Dunukofia Local Government Areas (L.G.As) of the Study area. The annual cumulative bio-wastes generated from the three LGA are 9,133,220 kg, 3,992,735 kg, and 6,909,450 kg respectively. It is recommended that government and stake-holders should take this study as a veritable tool in green energy technology implementation plan.

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1. Introduction

Globally, the need for renewable energy source(s) has been recognized as being efficient, economical and environmental friendly. The United Nations Sustainable Development Goal (SDG) number 7 is targeted at provision of affordable, reliable and modern energy services for everyone, it also aims at increasing substantially the share of energy mixed at regional and global level with renewable energy source. Most countries of the world have made efforts to achieve this goal through the development of biogas energy plants via thorough spatial analysis. One of the main

applications of GIS is in urban planning, urban planners use GIS both as a spatial database and as an analytical/modelling tool. Land suitability maps are very useful in the development of urban planning; they can be used to identify the solution space for future development. As global trend keeps advocating for a transition from fossil energy source to Renewable Energy (RE) source based on several socio-economic and environmental justifications, the necessity to embark on projects that utilizes biomass for energy production is inevitable for developing countries [1]. Biogas, which is bioenergy produced from biomass, has several advantages over other forms of renewable energies [2], hence the need to adopt and transit to biogas technology, particularly for oil rich countries in accordance to global trend.

It has been reported that the proper assessment of biogas potential is the first step in assessing the feasibility of establishing

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biogas-based power plants; this usually depends on the availability of the primary feedstocks. The availability of bio-wastes feedstock in a given area depends largely on the population of livestock and the density of farms; bio-waste sources could either be sparsely distributed or concentrated in a region. Therefore, information on the spatial distribution of livestock is crucial to assess the local availability of biogas feedstock [3]. Preliminary investigation of the various poultry farms and abattoirs in the study area shows that large volumes of bio-wastes are generated daily across farms and abattoirs in the State. The study further shows that poultry droppings is utilized as manure in only few farms from the months of April to July, while in non-farming season, the bio-waste is a nuisance and its disposal is done in such a manner that it constitutes serious environmental problems. The abattoir bio-waste similarly are allowed to accumulate in most abattoirs and disposed without considering the economic value; this has resulted to pollution of the environment and under-utilization of this rich energy source. The impact of these wastes on land and water emphasizes the need for appropriate strategies for efficient waste collection and processing in Nigeria's public abattoirs, farms, and environs generally.

Anaerobic digestion is gaining serious attention globally; a study by Ref. [3] emphasized the progress made in European Union to actualize bio-economy. The study reported that assessment of the spatial distribution of the biogas potential of farm manure from livestock and poultry is a key issue for the location and economic performances of a bioenergy plant; the authors computed biogas estimates through a spatial analysis algorithm that uses data of livestock and poultry manure production and collection. 1 km spatial resolution evaluation of the spatial distribution of biogas was achieved in the study. Brown et al. (2020) in another study opined that road-verge grass could serve as feedstock for energy generation in Lincolnshire area of UK. The study estimated the potential volumes and locations of road verge grass biomass, with capacities and locations of existing Anaerobic Digestion (AD) plants. The authors recommended that harvesting and co-digestion of verge grass would potentially reduce the cost of road verge maintenance and would increase biodiversity especially in the rural areas. This study shows that a variety of feedstock is being researched on, to ensure adequate provision of biomass for biogas plants. An investigation to assess the feasibility of converting biogas produced in an existing anaerobic digestion plant to bio-methane, was carried out by Baccioli et al. (2018), the study considered a biogas plant that is equipped with a micro-gas turbine co-generator. The most suitable configuration for system upgrade was considered, from thermodynamic and economic aspect. The study concluded that the feasibility depends on the amount of bio-methane generated, since this tends to affect the performance of the cogeneration system and the balance between the costs and revenues. These studies indicate the increasing attention on renewable energy adoption such as in feedstock identification and assessment, optimum location studies, feasibility studies on improving existing plants etc. for economic, environmental and technological benefits.

Effective planning of farm waste recycling programs is a substantial challenge for the current farm waste management systems in most developing countries. Siting bioenergy plants in optimal locations at optimum capacities is a challenging task [2]. Due to the high spatial dispersive nature of farms, implementation of spatial information technologies such as Remote Sensing and GIS in addressing this issue appears to be an appropriate methodology [4]. Siting of biogas plants in strategic locations is a major means of combating some of the environmental challenges of bio-waste generation; this would also be convenient and economically advantageous. Supply chain analysis is therefore undisputable in the realization of a sustainable waste management system. In order to

achieve this [5], embarked on a supply chain analytical research, using an innovative approach that integrates geospatial technology, set cover location modelling and location-allocation model for optimized locations considering spatial, socio-economic and technological factors. Despite the success of the work, there is need to consider several scenarios for the development of bioenergy plant locations for the State that would consider maximization of solar energy potential using the hill-shade layer map (this is geospatial 3D model of the earth surface, with the sun's relative position taken into account) for economic savings through plant location in proximity to the road; electric transmission lines, etc. This study therefore seeks to provide alternative suitable sites by employing more geospatial tools in GIS to exploit the potentials for adequate bio-resource mapping, which has been attributed to be a great tool in site analysis by several researchers [6]. Apart from the previous research on the supply chain analysis [7], recently conducted research on risk assessment of abattoirs located in proximity to water body in the municipal city of Onitsha, Nigeria, hence this study seeks to complement and provide elaborate geospatial analysis for the site selection challenge.

Several researchers have applied GIS as an appropriate tool for site suitability analysis [6,8]. In separate studies attempted to find out suitable urban land using GIS for the provision of urban amenities in Srinagar and Jammu, India and Beijing, China respectively. In Iran for instance Refs. [9], determined suitable sites for solar farm and wind turbine using GIS and Analytical Hierarchy Process (AHP) in Tehran, in order to generate a distributed network to increase power network stability. Similarly [10], developed a fuzzy multi-criteria decision making technique, integrated with GIS to assess suitable areas for rice cultivation in Amol District, Iran. In Portugal, GIS and Multi-Criteria Evaluation (MCE) techniques were used to assess the land suitability for the installation of solar farms in the municipality of Ourique in the South of Portugal by Ref. [11]. For bioenergy site suitability analysis, a number of research have been done. GIS supported methodology was developed in order to assess the technical and economic potential of biomass exploitation for energy production in Europe by Refs. [3]. [12] analyzed the spatial distribution and amount of potential biomass feedstock for bio-methane production and optimal locations, sizes and number of biogas plants in Southern Finland. These research works asserts that GIS is an appropriate and recognized spatial tool for location analysis. To the best of the author's knowledge, no research has been carried out in Anambra State that captures socio-economic and environmental criteria as done in this study for bioenergy site analysis using geospatial techniques; this of course will provide the State with enormous benefits in her recent efforts to adopt and transit to RE systems.

The government of Anambra state recently (2018 to be precise) contracted a foreign Australian company for the installation of bioenergy plant in the three major cities of the state viz: Awka, Onitsha and Nnewi. Preliminary investigations have shown that these areas were chosen arbitrarily without any location analysis. More also, emphasis by the government and previous researchers have focused mainly on municipal waste management, while agricultural waste management was relatively ignored [13]. There is therefore an urgent need to give attention to agricultural waste management through biogas technology in the State [14]. One of the obstacles to overcome in utilizing livestock bio-waste for bio-energy production in the study area, lies on the fact that the region's bio-wastes sources are scattered geospatially and there exist numerous subsistent agricultural holdings that generates the bio-waste (predominantly small and medium scale farms) [15]. recently asserted that to develop sustainable biogas system, the assessment of feedstocks availability is important, the authors reported that the application of GIS is a relevant tool for investigating

both biomass availability and sustainable biogas plant locations. Despite the fact that GIS site suitability studies on biogas plants have been done by various researchers in the developed countries, little or few studies have applied GIS for bioenergy development in many developing countries. There is need for researchers in developing countries (like Nigeria) to embark on similar studies to ensure universal transition to renewable energy sources.

It is worthy to note that various researchers from different regions of the world developed their site suitability studies based on their geographical peculiarities. On the novelty for this study, it is worthy to note for instance, that previous studies on biogas plant suitability analysis conducted by researchers from developed countries (mostly from temperate regions) have failed to incorporate hill-shade view thematic layer in their site analysis, probably because of the low impact or contribution of solar radiation to biogas production in these regions when compared to the tropics, this is understandable. Consequently, there is need for site suitability analysis as done in this study, to take into cognizance the geographical peculiarity of the tropical region. The inclusion of hill-shade view will invariably provide profound benefit in site selection for the region. To the best of the authors' knowledge, this has not been applied in biogas plant site selection analysis. Furthermore, similar studies on biogas plant suitability analysis do not exist for Nigeria. In addition, since most developing countries are gradually giving attention towards transition to renewable energy source, this study will improve the current approach for the location of biogas facilities. Hence, this study is an attempt to provide exploratory study that will specifically present a logical framework for guiding the process of identifying potentially suitable sites for biogas plant in developing countries using the study area as a case study.

2. Materials and method

2.1. Study area

Anambra State (with capital of the State situated in Awka) is one of the most populous states in Nigeria; the State occupies land mass approximated to be about 4844 square kilometers. According to the national population census of 2006, the population of the State is estimated to be about 4.06 million, with a population density of 1500 to 2000 persons living within every square kilometer. The topography and climatic vegetation of Anambra State of Nigeria makes the area very suitable for extensive modern agricultural investment. It is endowed with 60% arable land which should be for large-scale farming. The climate of the study area is the tropical rainforest, comprising of two seasons; the rainy and dry seasons. The rainy season lasts from March to October, while the dry season lasts from November to February. The annual rainfall is about 1300 mm, peak rainfall occurs in June to July while the second peak occurs in September to October, the two wet seasons are normally separated by a drought (August-break). The rains could be mild or torrential and often causes flooding and erosion leading to the formation of gullies. Data from the Federal Ministry of Aviation, Meteorological Department, Awka as reported in Anambra statistical year book of 2009 shows that the State experiences the hottest temperature in the month of March and the lowest minimum temperature in the month of December. Annual average temperature ranges from a minimum of 20.78 °C to a maximum of 30.24 °C. Relative humidity ranges from annual minimum of 34% in dry season to a maximum of 89% in rainy season. The map of Anambra State is shown in Fig. 1.

There are many poultry farms in the region, few pig farms and nomadic rearing of cattle. For this study, the bio-waste investigated was limited to poultry droppings and abattoir wastes because of its

abundance and availability. The pig dropping was not considered because of non-availability of data. Preliminary studies also show that because of the nomadic nature of cattle rearing in the region, accessibility of the waste is difficult and not readily available for collection.

2.2. Data collection and sources

This study focuses on site suitability analysis of Anambra State of Nigeria for biogas siting using GIS techniques. The primary data was obtained through field survey; the data were collected through visits to Agricultural livestock farms and slaughter houses in the study area, to determine the biomass potentials of the various farms and abattoirs. A Global Positioning System (GPS) receiver was used in field survey for the ground-truthing and to determine the geographical co-ordinate of the farms for geo-coding in the GIS data analysis. To determine the total bio-waste generated from the various agricultural waste generating points of the study area, estimate of the manure excreted by 1000 birds per day was approximated to 120 kg for layer chickens, 80 kg for meat chickens, 200 kg for turkeys and 150 kg for ducks [16]. More than 2000 poultry registered farms, with about 1,844,557 birds were used in the data analysis as reported by Anambra State Veterinary Department. The total wastes collected from the various livestock centers in the community were summed up, and the towns analyzed as a point data for simplicity, the waste generated was used to develop the bio-waste thematic layer or the bio-waste spatial density map, this was obtained from point density tool in spatial density toolbox in ArcGIS. The spatial density analysis of biomass availability takes known quantities of agricultural wastes obtained from field survey and spreads them across the landscape based on the quantity that is measured at each location and the spatial relationship of the locations of the measured quantities. About 115 communities were presented as bio-waste generating source; this is shown in Appendix 1 with the corresponding annual bio-waste generated. Data were also collected from Ministry of Land and Survey, Awka, which include boundary map and administrative land use map. The GIS-based thematic maps used for the production of the suitability map include the political boundary map, road network, LULC, DEM, river, slope, biomass potential density layers etc. The flowchart of the methodology is as shown in Fig. 2:

2.3. ' data processing and data analysis

The study was undertaken with a pre-processed and corrected Landsat-7 ETM + image, this was geographically registered before being used for the analysis. The imagery was processed at the National Centre for Remote Sensing, Jos, Nigeria. The radiometric data corrections were effected on the Landsat-7 ETM + imagery, the important radiometric corrections implemented include: internal calibration, atmospheric corrections, corrections for sun elevation, topography etc. A composite image was generated from the Landsat-7 ETM + image and then classified to extract the different land uses of the study area using supervised classification algorithm provided by ArcGIS 10 software. Slope layer was derived from the DEM of the study area; the layer was obtained by spatial analysis using the slope function in Spatial Analyst Tool in ArcGIS. The thematic maps were prepared and edited, overlaid and visualized on the basis of the suitability analysis using ArcGIS 10 software of ESRI. The application of GIS for overlaying thematic layers to establish land databases requires that all the layer maps be converted into a common coordinate system of WGS 84 ellipsoid. This involves a stepwise arrangement and organization of acquired data in a manner that is appropriate for analysis. Environmental, social and economic factors were considered in the GIS overlay analysis

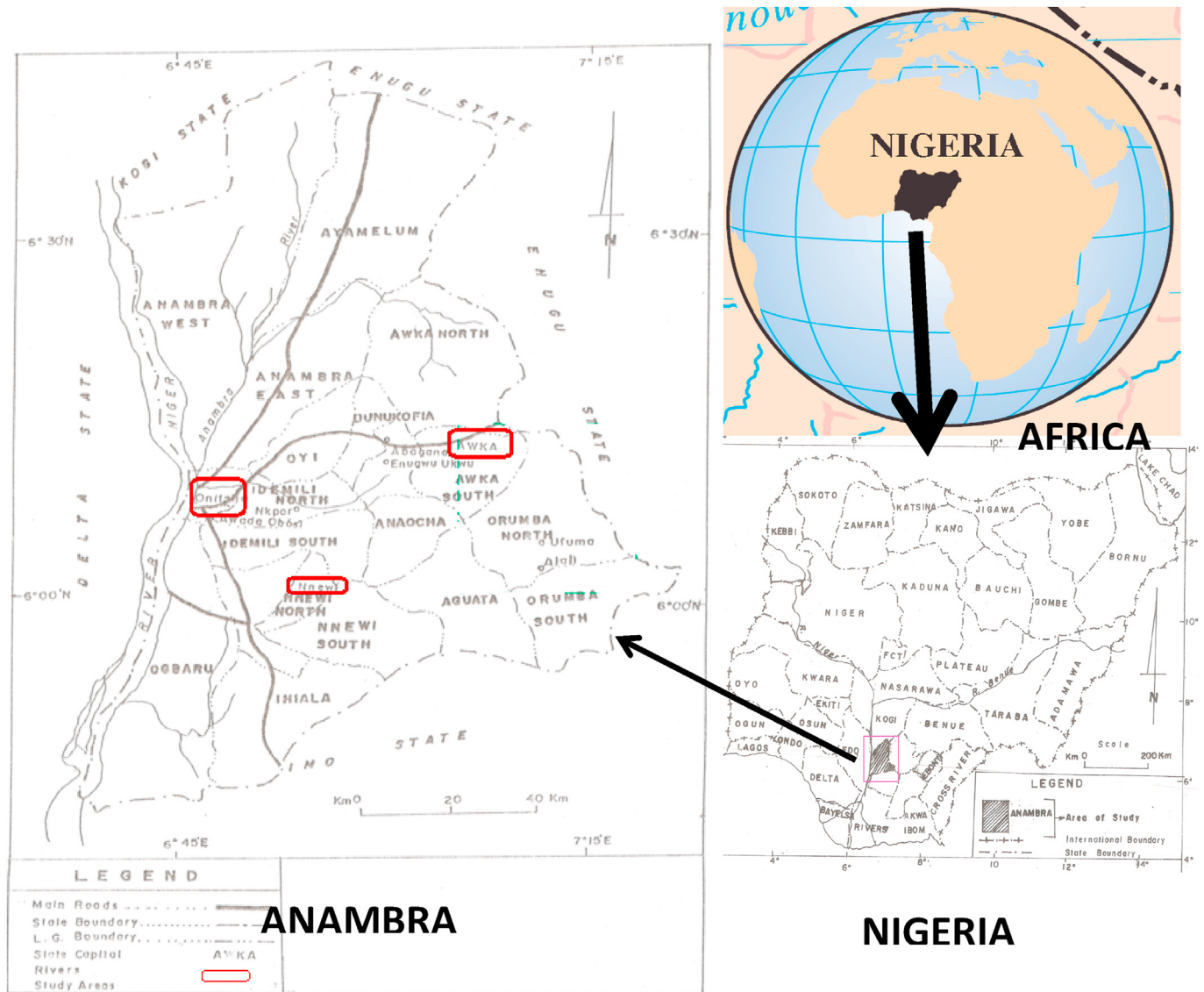


Fig. 1. Map of the study area with major cities highlighted.

for the production of the final suitability map. The gross electrical power P_e (kWe) of a biogas plant can be expressed as a function of the input biomass according to the formula [17]:

$$P_e = (Q_p \times TS_p \times VS_p \times BY_p \times BA_p) \frac{CH_4 \times LHV \times \eta_e}{H} \quad 1$$

Where Q_p (t/yr) is the potential annual livestock manure available for digestion, TS (20%) is the Total Solid, VS (80%) is the Volatile Solid, BY is the biogas yield (60%, $N\ m^3/t$) and BA is the Biogas Availability of the biomass (80%); CH_4 (60%) is the percentage of natural gas in the biogas and LHV (25.2 $kW\ h/Nm^3$) is the Low Heating Value of natural gas; η_e (44%) is the electric efficiency of the power plant and H the annual operating hours (7468 h/yr). Equation (1) was used to determine the gross electrical potential of the bio-waste in various communities; this is shown in Appendix 1. This is also related to recent report on spatial analysis of biogas potential from manure in Europe by Ref. [3].

2.4. Site selection procedure

To identify areas with high biogas potential which will eventually represent suitable areas for bioenergy siting considering transportation cost and raw material availability; the biogas production potential was analyzed using livestock (which includes poultry wastes and abattoir) waste generation capacity of the study area. The flowchart of this process is shown in Fig. 3; four crucial steps were used to produce site suitability map for bioenergy plant and these are: (1) finding suitable factors to be used in the analysis, (2) generating land suitability maps by using various criteria, (3) assigning factor priority and class weight (rating) to the parameters involved, and (4) determining potential areas for bioenergy plant.

The economic principle applied in this study is based on the theory that profit maximization can only be achieved by cost minimization. The economic factors considered includes proximity to electric transmission line, access road and biomass source. Though the solar-based heating against artificial heating of the biogas plant, will lead to energy savings, it was applied under the land use suitability map since it is associated with land use. It

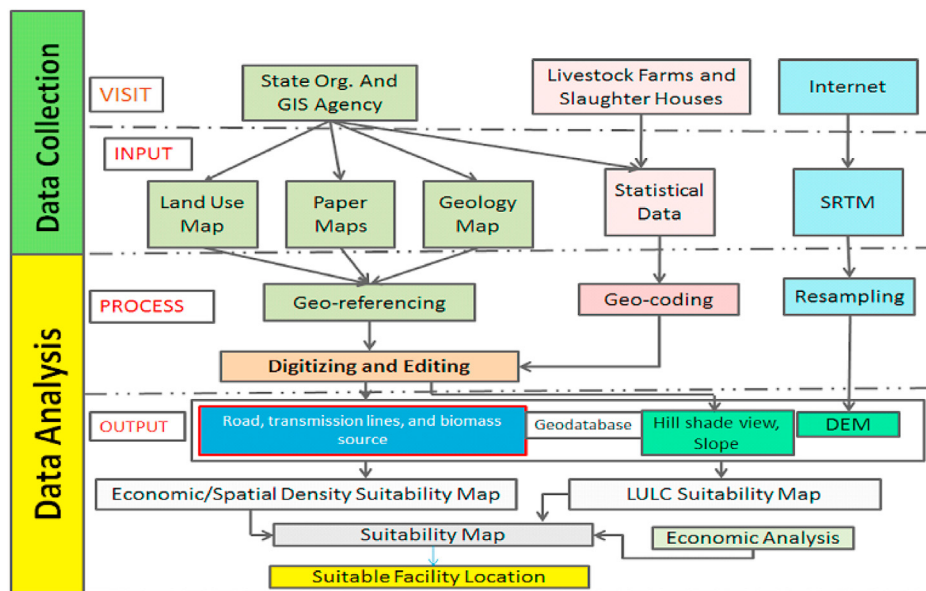


Fig. 2. Flowchart of methodology.

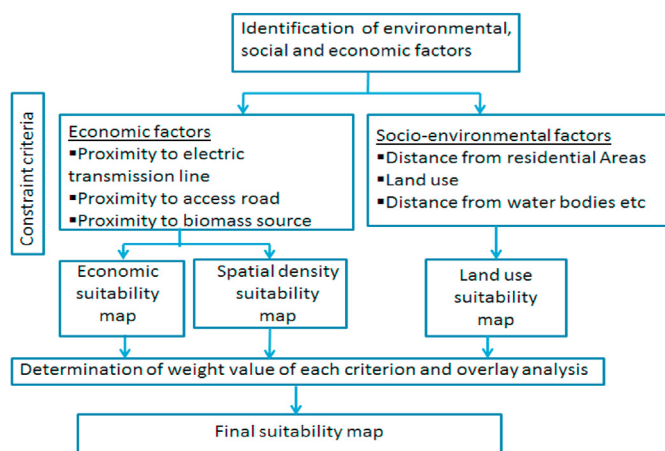


Fig. 3. Schematic diagram for modelling suitable bioenergy plant sites.

implies that these economic factors should be minimized to increase profitability, with the exception of solar heating that will be maximized for increased profitability.

Spatial modelling which involves applying GIS functions to some spatial data was undertaken. This includes geometric modelling functions: generating buffers is a type of geometric modelling analysis, and is one of the spatial models in modelling the bioenergy site. Buffer is a type of GIS analysis of finding what is near a feature. One way to find what is near a feature is by creating a buffer around the feature. GIS site suitability analysis could identify suitable site within a given distance of a proposed bioenergy site by buffering, for instance, the economic dependent features. The buffer could be used with other layers of data to show which sites would be near the bioenergy centre. It is proposed here that for economic reasons, the bioenergy centre should be located preferably within 100 m of the road network. The farther it is, the less economically viable the site is. A buffer of 100 m, 1 km, 3 km and 5 km were created. Another economic factor in bioenergy siting is distance to electric transmission line, transmission losses can be reduced by siting bioenergy facilities close to transmission lines

[18], considered distance to transmission line in their study, a buffer of 1 km, 3 km, 5 km and 10 km was therefore created for this. Since agricultural wastes is the major source of waste for the bioenergy plant for this study area, the driving distance of 5 km, 10 km, 15 km and 20 km was used for all the communities generating bio-wastes. As economic factor, road was given a weightage of 85% and electric transmission line 15% to produce economic suitability map. The proximity of the plant to road network was given higher weightage more than the transmission line because the transmission line constitute a one-time investment (fixed cost) in grid line installation, while the road involves continuous operational investment (variable cost) of loading and unloading of the bio-waste to the plant, this is essential in reduction of logistical and operational cost of the plant, this is in accordance to previous studies [19,20]. The slope layer, hill shade layer and DEM layers were overlaid at 20%, 40% and 40% weight respectively using weighted overlay tool to produce the land use map, while the bio-waste map constituted the bio-resource spatial density map. The economic and spatial density maps were overlaid at 50% weightage respectively to obtain the economic/spatial density suitability map, similarly the land use suitability map and economic/spatial density suitability map were overlaid at 50% weight respectively to obtain the final suitability map. Constraint map was developed for the analysis, the constraint map criteria are shown in Table 1. The map was based on specifications of previous researchers [12]. The weighted overlay tool in ArcGIS was used to overlay the raster dataset using a common measurement scale and weights according to its importance. Suitable areas for AD were evaluated to avoid close proximity to land features and uses that may be sensitive to the characteristics of utility-scale power production including surface water, wetlands, forests, public lands, highly sloped lands, and developed residential areas, with acceptable slopes of 14° or fewer as suggested by Ref. [21].

3. Result and discussion

The result of the study was the derivation and overlaying of the various thematic layers to determine the biogas plant suitable site. The results of the various thematic layers used in the biogas plant suitability analysis are discussed below:

Table 1
Constraint map criteria.

Constraints	Specifications
Rural and urban areas	A distance of 1 km away from residential and urban areas
Park and recreational areas	Sites falling within these areas and a buffer of 500 m are avoided
Rivers, lakes and other waterbodies	Sites within a buffer zone of 200 m are avoided
Wetlands	Wetland areas and a buffer zone of 200 m are avoided
Environmentally sensitive areas (flood plains, conservation areas, habitat sites)	Sites falling within such areas and a buffer zone of 500 m are avoided
Roads	Sites falling within a buffer of 30 m are avoided
Transmission line	Sites falling within a buffer of 100 m are avoided
Land surface gradient	Areas with slopes larger than 15% are avoided

3.1. Road network and river thematic layer

The major road networks were digitalized and prepared to ensure that transportation cost is minimized. The thematic layer for the road network was represented by polylines as shown in Fig. 4a. This shows that Anambra West and Ogbaru L.G.A has the least number of road networks, indicating that the possibility of collecting bio-waste from the various waste generation sites in these regions would be difficult and economically disadvantageous in comparison to other regions with greater number of road networks. Considering proximity of siting biogas plant to road networks, these areas would also be unsuitable. A lot of road network is seen in the central region of the State, which would also enhance bio-waste collection from the dispersed waste generating sources, and could also provide economic advantage for siting of biogas plant. Fig. 4b shows the digitized river shapefile. River Niger cuts across the State boundary towards the West. Tributary rivers from various towns could also be seen in this Figure, cutting across several towns. Majority of the river network is seen in Anambra West located in the Northern part of the State, this also makes the region unsuitable for location of biogas plants, to avoid possible contamination of the water bodies.

3.2. Electric transmission line thematic layer

Electricity is generated at a power station (coal, natural gas, nuclear plants, wind turbines, hydro and solar power), stepped up to a high voltage for long distance transportation over transmission lines, and then stepped down again at a substation. The electric utility companies' transmission and distribution system links power plants to customers through high power transmission line network, the shapefile of electric transmission line distribution of the study area is shown in Fig. 5:

The electric transmission line provides the layer that ensures that sites for bioenergy installation are located in proximity to powerline; this will reduce energy losses in the electric transmission process. It is expected that suitable locations should be in proximity to these lines to save both energy and to reduce the cost of transmitting energy generated to the grid. The Figure shows that the network of electric transmission lines runs through various cities, these lines were terminated in nearby states (Enugu State and Imo State mainly). All the 21 Local Government Areas (LGA) were transverse by these electric transmission lines, however the central region shows higher transverse density of these lines. The Northern region shows few lines and therefore has low potential

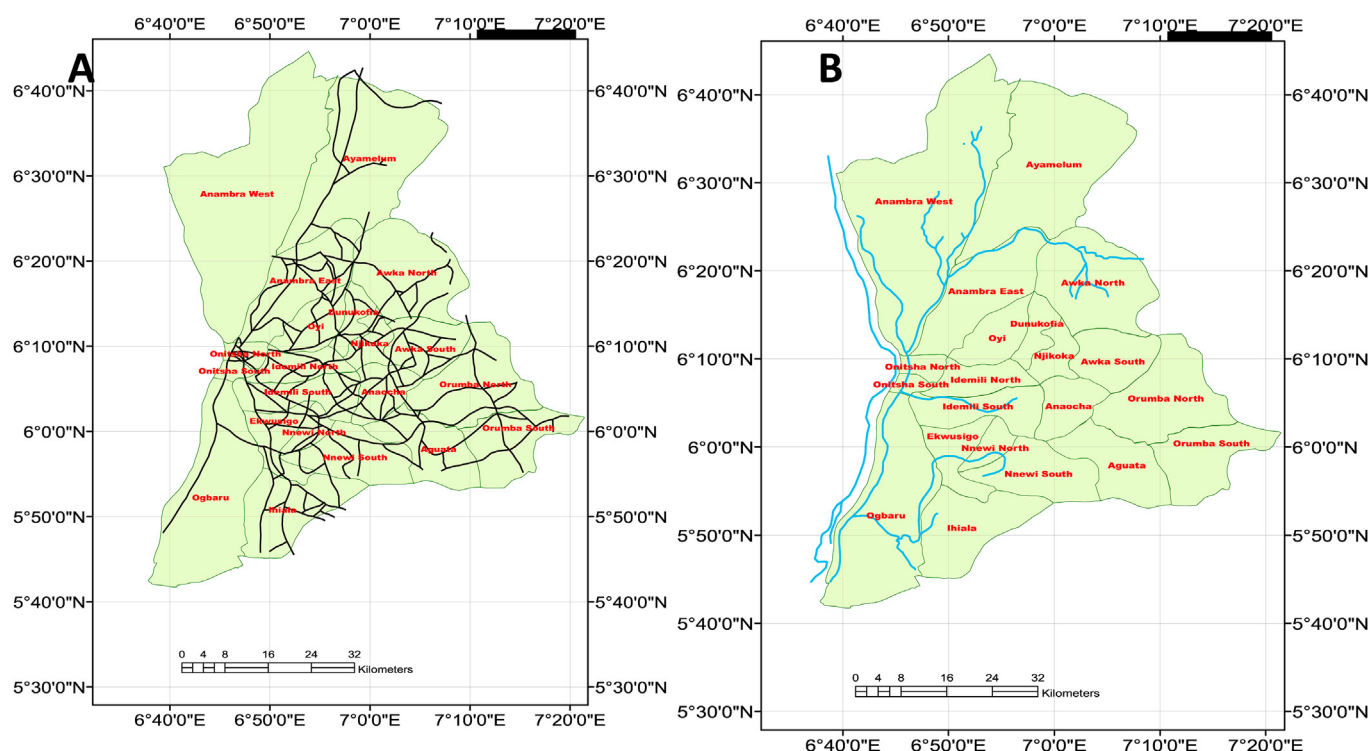


Fig. 4. (a) Road and (b) river shapefile of anambra state.

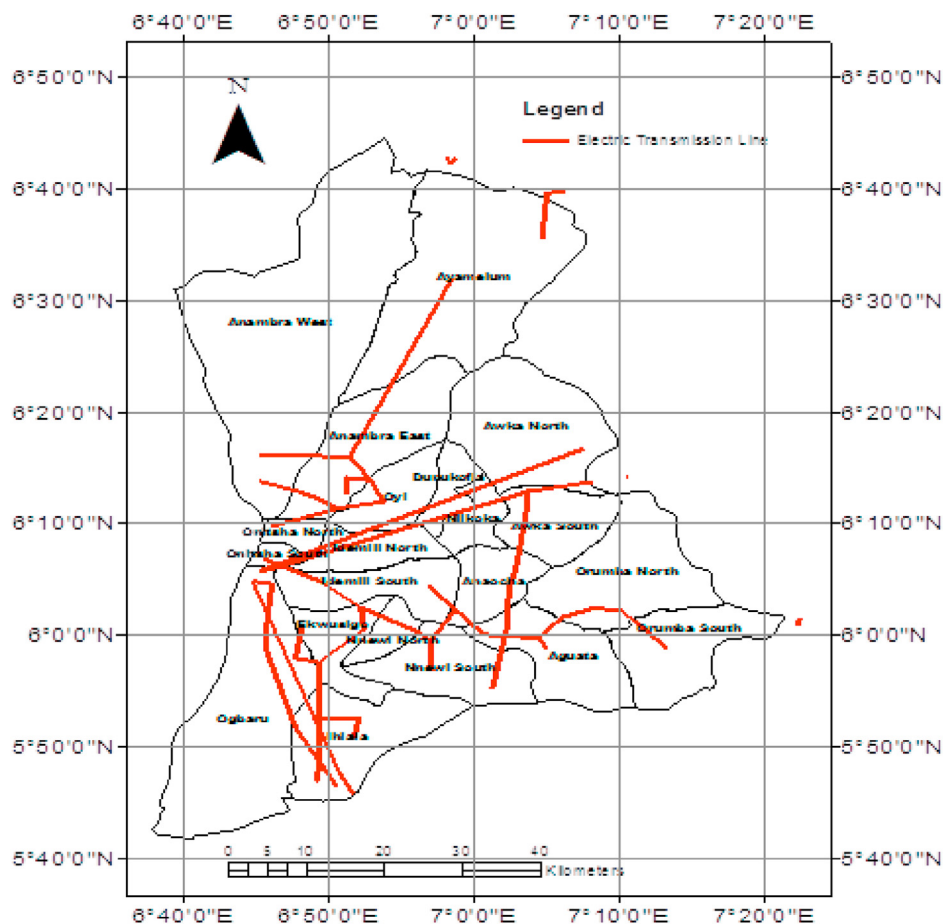


Fig. 5. Shapefile of Power Plant and Transmission lines in Anambra State.

for installation of bioenergy plants. The region with relative high transverse of the transmission lines would be more economical for siting biogas plants. It is therefore inferred that the Central and Southern regions would be optimal location for biogas plant siting in consideration of proximity to electric transmission network.

3.3. Shuttle radar topography mission (SRTM) and hill-shade view

The SRTM Image was in Latitude, Longitude coordinate frame of WGS 84 ellipsoid. The Calculate Tool in raster properties in ArcGIS was used to resample the SRTM data. The SRTM Image was clipped for Anambra Administrative area. This is shown in Fig. 6 (a). The SRTM Imagery was the source of elevation points for this study and was used to produce contour of the study area. The elevation information shows that the elevation of the study area ranges from 0 to 385 m.

From the Figure, the white colored areas represent areas in the region with the highest elevation, the white colored area are seen mostly in the Southern region. The West, West-north and West south areas, characterized with low elevation are unsuitable for siting bioenergy plants. Preliminary studies show that these areas are closer to the River Niger, and are majorly wetlands; they do experience flooding during certain period of rainfall over the years. The Central regions of the study area are characterized with high elevation (designated mainly with chocolate color) with few areas in white color. This region could serve as suitable sites for bioenergy plants in consideration of high elevation criteria. The hill-shade view layer is shown in Fig. 6 (b), it is a grayscale 3D model of the

earth surface, with the sun's relative position taken into account for shading in the image. The hill-shade Figure has an azimuth of 315° and an altitude of 45°. This function uses the latitude and azimuth properties to specify the sun's position. Biogas production is known to be dependent on temperature, the biogas plant then should be situated for better exposure to solar energy, since biogas production is dependent on the heat energy, the best site for location of bioenergy plant should be on a site with maximum solar heat source for heating the biogas plant. The hill-shade view value ranged from 0 to 181. A critical look at the Figure shows that there is higher solar radiation at the Central region of the state; this area also seems to have high elevation. The dark areas that represent shaded areas will not be a better option for biogas plant installation because of reduced solar energy exposure. To maximize the naturally available solar energy, consideration for installations of bioenergy plants should be in areas with optimal hill-shade properties, these is predominant at the Central region.

3.4. Slope factor and spatial density of biomass resource

Suitable areas for AD are best positioned in areas with acceptable slopes of 14° or fewer. The slope of the study area ranges from 0 to 89° degree. The slope obtained in the GIS operations was divided into 5 intervals to enable classification into suitable criteria and subsequent GIS overlay for the overall land suitability analysis. The slope raster dataset used for land suitability is shown in Fig. 7a:

The Figure shows that greater percentage of the study area is mostly characterized with high degree slope. It is worthy to note

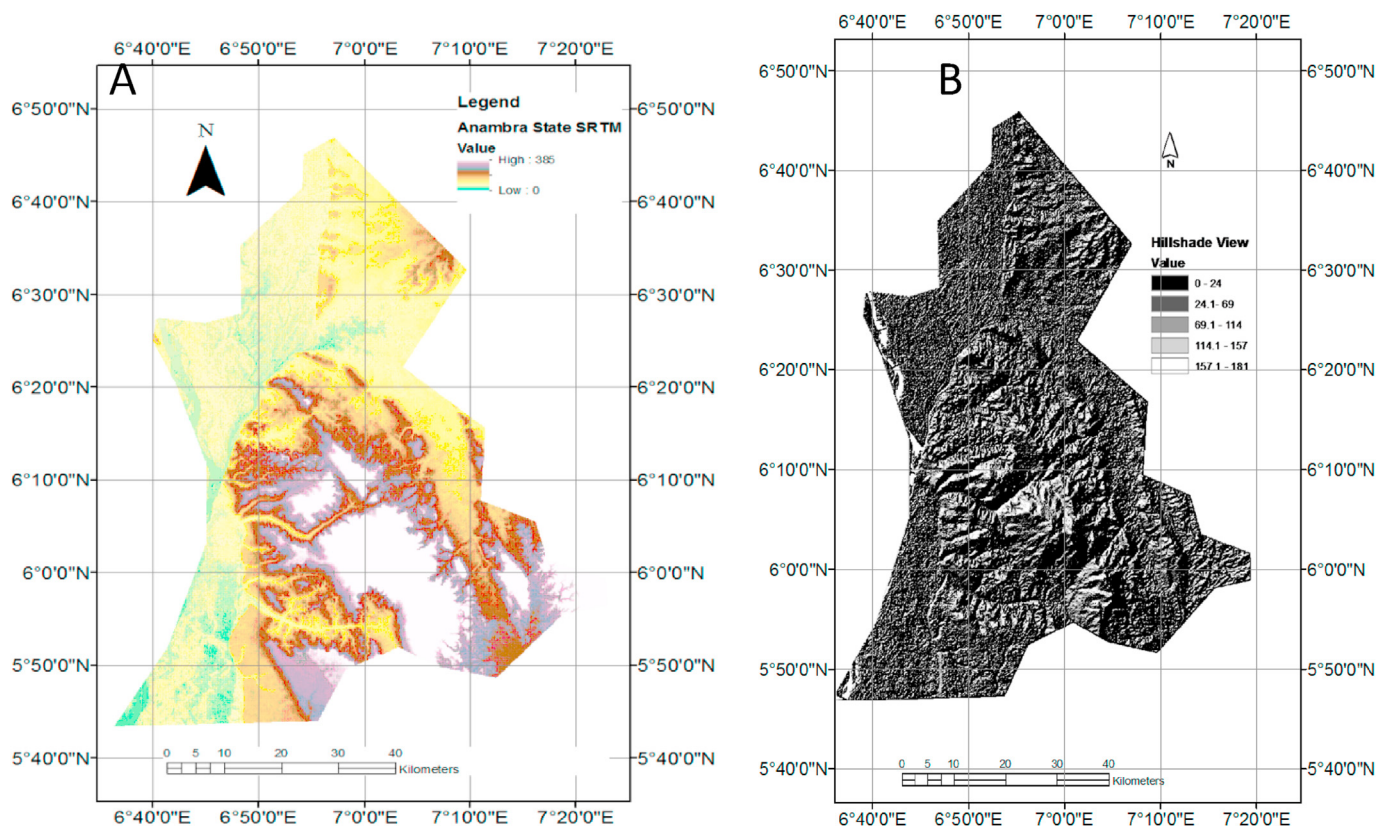


Fig. 6. (a) Anambra SRTM and (b) Hillshade view layers.

that the areas mostly classified as suitable in the elevation map (SRTM) appears to be unsuitable in the slope layer, for instance, the white colored areas in the SRTM map that are seen mostly in the South-east region; with the highest elevation is represented in this Figure as unsuitable. Also the West, West-north and West-south areas characterized with low elevation and were therefore unsuitable based on elevation criteria are classified as suitable areas based on the slope factor in siting bioenergy plants. The ability to combine and utilize these contrasting geospatial features has been proven to be one of the advantages of GIS in location modelling. However, patches of most suitable sites are found in all the areas, but are predominant in the West and Northern region. To map the biomass resource, the spatial density and dispersiveness of the various bio-waste communities is essential and is shown in Fig. 7b. Bio-waste production potential, availability and spatial density is one of the factors considered in biogas plant installation. From the Figure, Anambra West L.G.A has no bio-waste generating communities; this has been attributed to the swampy nature of the area. Hence, the region was eliminated in the study area for siting biogas plant. Ayamelum L.G.A has only three bio-waste production communities, which include Anaku, Omor and Ifite-Ogwari. From the Figure Omor communities has the highest waste generating capacity from the L.G.A. Several bio-waste production communities in a Local government indicate higher biomass availability/accessibility and consequently available sites suitable for locating bio-energy plants. The Figure also shows that Ogbaru town has the highest point density which is above 4,362,480 kg of poultry droppings annually. This is followed by Umuchu, Onitsha, Nise, Nibo and Amawbia, with point density that ranges from 2,210,250 kg to 4,362,480 kg.

3.5. Economic and spatial density suitability analysis

Economic factors were determined as a raster dataset, also the spatial dependent factors were integrated as raster dataset, both raster dataset were integrated to produce the economic and spatial density suitability map as detailed below: The two economic factors considered are proximity to road and proximity to electric transmission line. Since proximity to road is considered to incur more cost as a result of it constituting a variable cost, more weight was given to this factor (85%) while proximity to electric transmission line as a fixed cost was given a weighted value of 15%, this is comparable to the work of [4]; where road network was given preference than transmission line in a pair-wise comparison matrix and weights of preference factors in AHP. Fig. 8 shows that areas within the 100 m and 1 km buffer of road network were indicated as most and highly suitable sites respectively.

3 km of the road network was indicated generally as moderately suitable sites while 5 km distance from the road network was generally classified as less suitable. From the combination of the economic and spatial density factor in determination of suitable sites for location of bioenergy plant, the most suitable areas were found in Onitsha North and Njikoka L.G.A. The highly suitable locations are within Onitsha North and South, Njikoka, Idemili, Egwuiso L.G.As. The moderately suitable sites are located at the extreme of Onitsha North and parts of Onitsha South; 100 m along Anambra East and Ayamelum L.G.A major road; along the expressway connecting Oyi, Dunukofia, Awka North and South L.G.A. Other moderately suitable sites include Idemili South, Ogbaru, Egwusigo, Nnewi North and Ihiala, Orumba North/South, and Idemili North. L.G.As, other areas shown in the Figure were classified as either less suitable or not suitable.

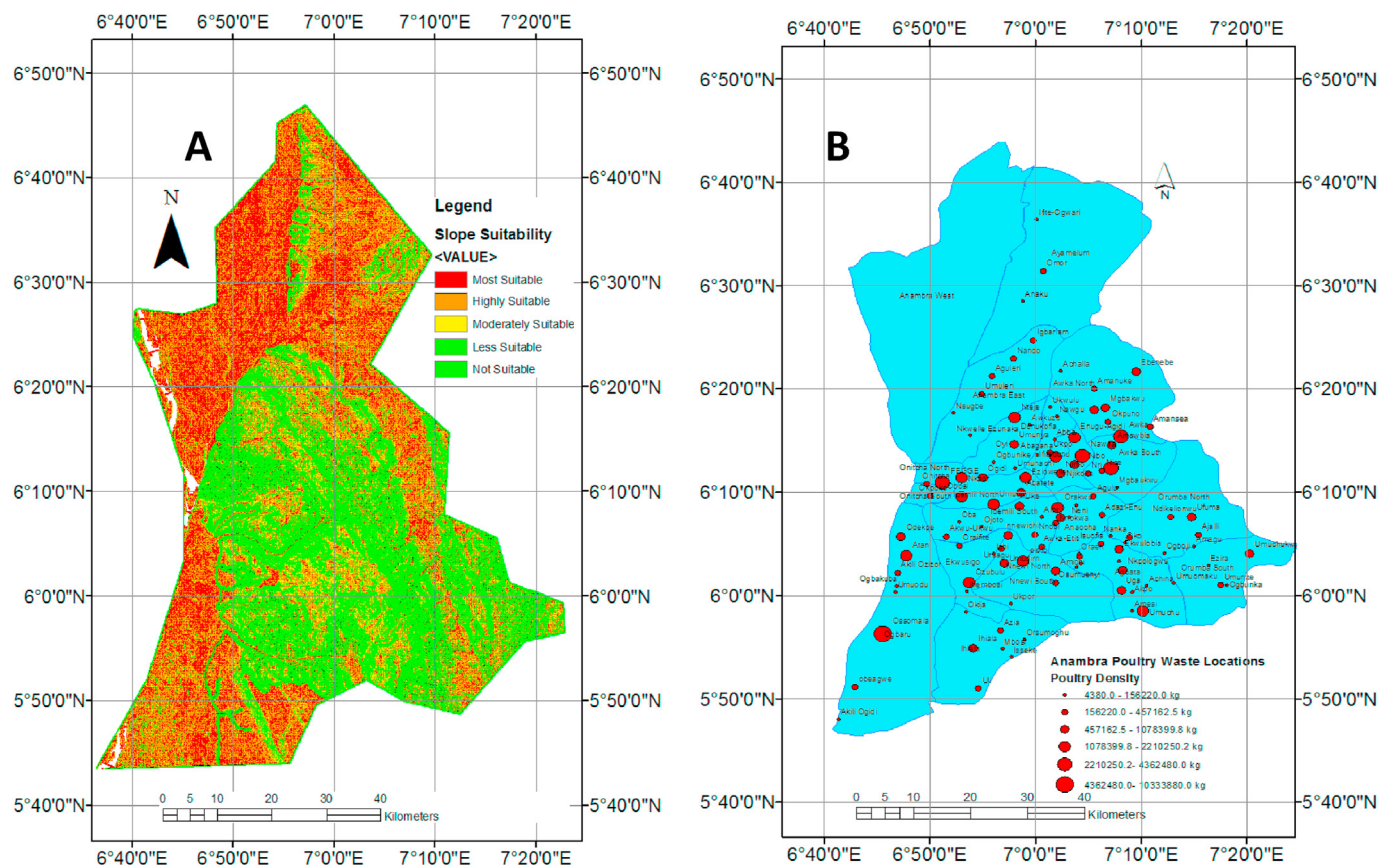


Fig. 7. (a) Slope and (b) Bio-waste spatial density layer.

3.6. Land use suitability map

The land use suitability map was obtained by first developing the constraint map by excluding unwanted areas identified in the constraint criteria as shown in Table 1. These are areas prone to flooding, areas that are close to residential areas, water bodies etc. The slope, hill-shade view and elevation layers were overlay at 20%, 40% and 40% weight respectively using weighted overlay tool to produce the land suitability map. The resultant raster is shown in Fig. 9. The Land-use suitability analysis shown is graduated from the “Most Suitable” to the “Not Suitable” sites.

The land suitability map suggests that the most suitable areas are located mainly in Aguata, Aniocha, Orumba South/North, Nnewi South and North L.G.As. Ogbaru L.G.A was majorly classified as moderately suitable area because of its low elevation which makes the area unsuitable for siting bioenergy plants considering the risk to flooding. The areas delineated as “Most Suitable” sites are areas with several thematic map advantages which includes high elevation, good solar radiation exposure, good slope factor etc. The integration of these good land features as modelled above is essential in bioenergy location modelling: for provision of adequate solar energy, protection from flood, savings in the bioenergy plant construction as a result of good slope etc.

3.7. Final suitability map

The final GIS-based biogas plant suitability map is the resultant of the integration (through geospatial overlay) of land use

suitability map, economic and spatial density suitability maps. This is shown in Fig. 10.

A similar study on site suitability analysis was conducted by Ref. [22] in Thailand, the study seek to identify ideal sites to locate in Songkhla, a province in southern Thailand, utility-scale wind and solar farms. GIS and AHP was equally utilized in the study, however they used physiographic, environmental and economic siting criteria in the site analysis. This shows similarity in the use of economic and environmental criteria as also used in this study, however, the differences lies in the use of physiographic criteria. Obviously, the objective of the suitability analysis and parameters that is associated with the objectives are guiding principles on criteria to be used. The final suitability map as shown indicates patches of sites designated as most suitable and highly suitable sites etc. About 186 polygons were identified as the most suitable sites; they were located in Njikoka, Onitsha North and Dunukofia L.G.As. This corresponds to the fact that several thematic maps indicated that these most suitable sites have comparative advantages over others such as been in proximity to electric transmission line, road network, high biomass availability and possession of suitable land features. Fig. 10 (b) shows some of the most suitable sites in Njikoka L.G.A., the Figure shows a number of most suitable polygon sites, these sites were intersected by a major road from Onitsha to Awka. The Figure also shows the buffered electric transmission line across the major road, indicating that the proximity of major roads and electric transmission lines was achieved in the site selection through GIS analysis. Other highly suitable sites shown could be harnessed for siting bioenergy plant should the available most suitable sites be insufficient or rejected for any reason.

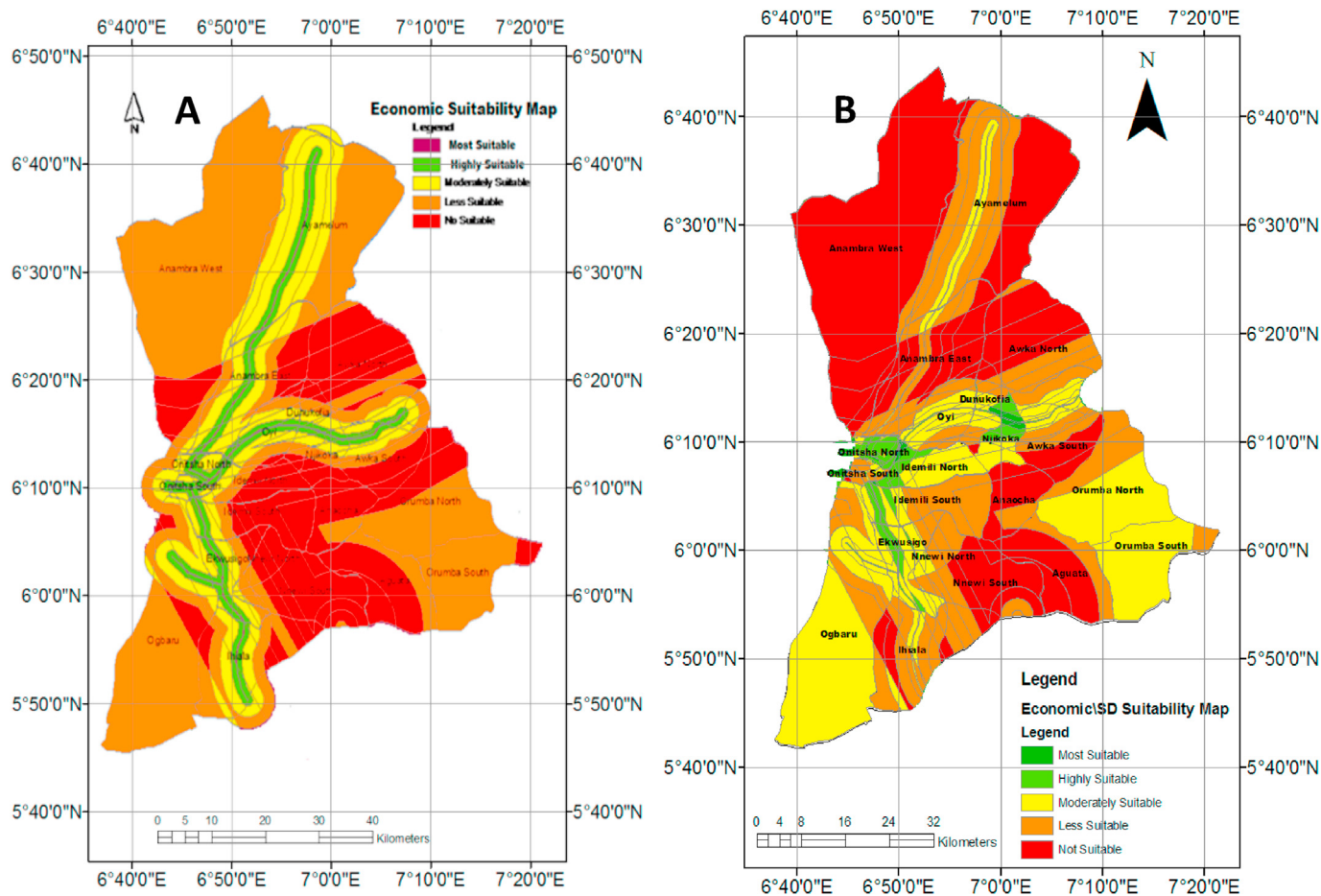


Fig. 8. (a) Economic suitability map and (b) economic and spatial density suitability map.

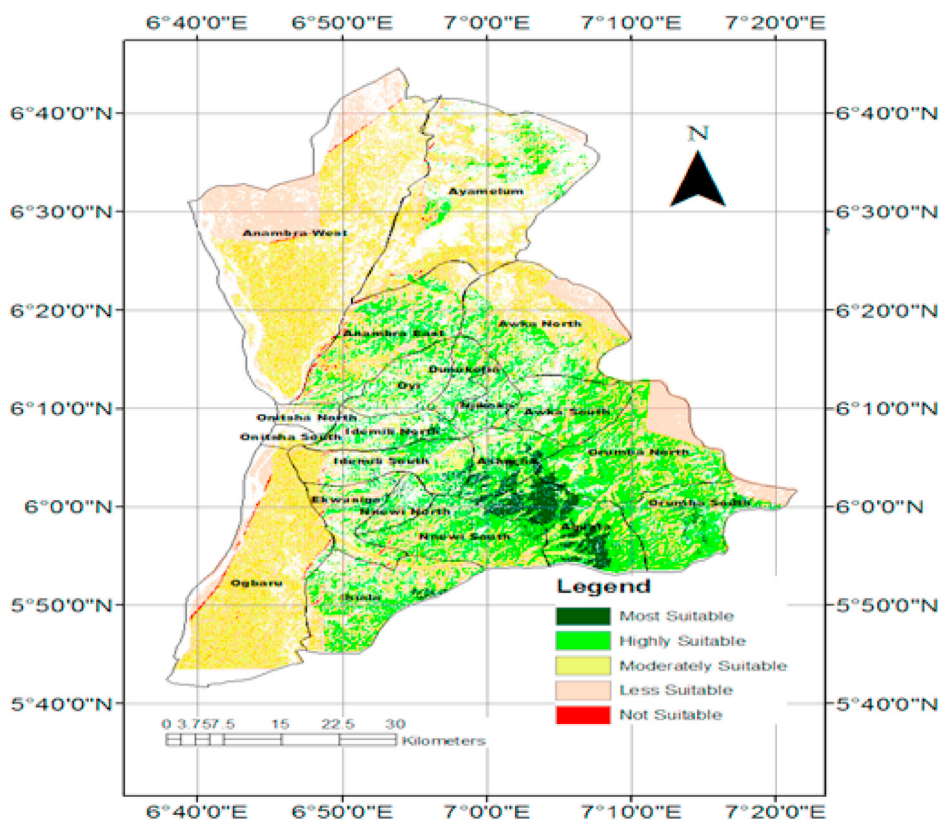


Fig. 9. Land use suitability map.

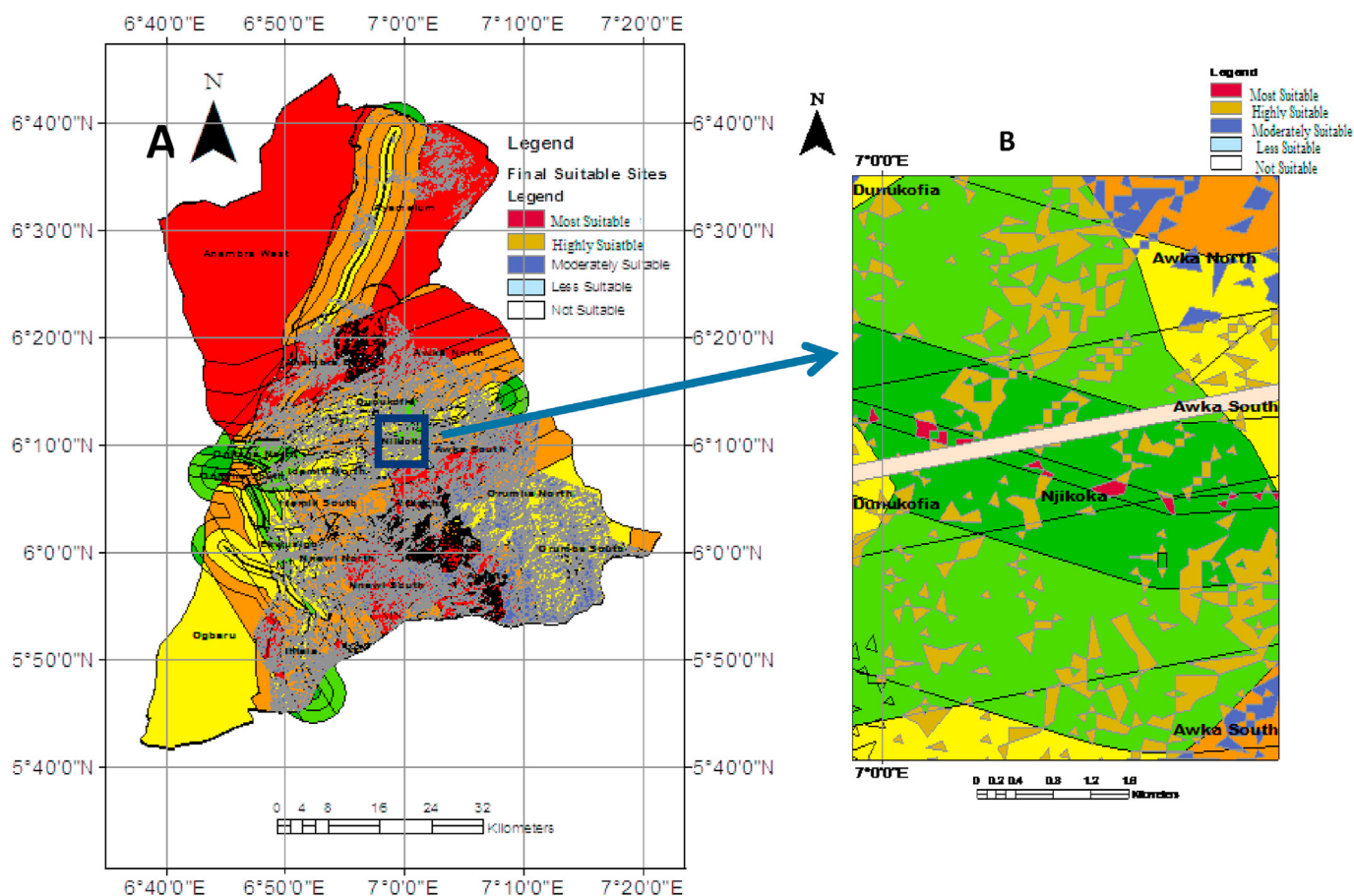


Fig. 10. (a) Final Suitability Map of the Study Area (b) Njikoka L.G.A showing some Most Suitable sites.

4. Conclusion

Consequent on the need to transit from fossil energy system to renewable energy system, this study was undertaken to determine suitable location for bioenergy plant using the power of GIS technology. Several GIS thematic layers were developed in the bioenergy plant location analysis. Data were collected for the GIS thematic maps from the Ministry of Land and Survey, Awka. Live-stock estimate was made through onsite visits to various livestock centers and from relevant ministries. Several factors were considered in the modelling process: socio-economic, environmental and biomass availability factors. The economic factors considered proximity to electric transmission line, access road and biomass source. The social and environmental criteria were used in modelling the constrain map based on identified social and environmental criteria. These factors were used to develop the economic, spatial density suitability, and land use maps. The weighted overlay tool in ArcGIS was used to overlay these raster dataset using a common measurement scale and weights according to its importance. The result of integrating these GIS thematic maps is a biogas plant suitability map, from the map, the most suitable sites to locate the bioenergy plants were identified in Njikoka, Onitsha North and Dunukofia L.G.As of the Study area. Systematic studies such as this is veritable in renewable energy planning and should guide stakeholders in policy making and decisions. This study was limited to potential energy supply from bio-waste generated within the region, further research should focus on potential energy demand from the region: this is usually dependent on the population of the various communities and their energy consumptive use. Also,

future research should consider time series data analysis of biomass generation, this will provide insight on biomass fluctuation or dynamic spatio-temporal aspects of biomass supply to biogas plant. Proper assessment of these variables will ensure that renewable energy mix as proposed in this study matches with the energy demand.

CRediT authorship contribution statement

Emmanuel Chibundo Chukwuma: Conceptualization, Methodology, Writing - original draft, Supervision. **Faith Chinenye Okey-Onyesolu:** Writing - review & editing, Software. **Kingsley Amaechi Ani:** Writing - review & editing, Software. **Emmanuel Chukwudi Nwanna:** Data curation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.renene.2020.09.046>.

Appendix 1. Bio-waste producing communities and bioenergy potential

S/N	Community	Waste/day	P (kg/year)	Biogas Potential (kWe)	S/N	Community	Waste/day	P (kg/year)	Biogas Potential (kWe)
AGUATA L.G.A					58	Nanka	426	155,490	10.64
1	Achina	240	87,600	5.99	59	Ndikelionwu	1014	370,110	25.32
2	Akpo	48	17,520	0.96	60	Oko	208.2	75,993	5.20
3	Amesi	428	156,220	11	61	Ufuma	1518	554,070	37.91
4	Ekwulobia	1501	547,683	37.47	ORUMBA SOUTH LGA				
5	Nkpologwu	1556	567,940	38.86	62	Ezira	222	81,030	5.54
6	Oraeri	288	105,120	7.19	63	Ogbunka	144	52,560	3.60
7	Ifite-Ezinifite	152	55,480	3.80	64	Umuchukwu	1602	584,730	40.01
8	Agulu-Ezechukwu	55	20,075	1.37	65	Umunze	986.4	360,036	24.63
9	Igboukwu	770	281,229	19.24	66	Umuomaku	132	48,180	3.30
10	Isuofia	488	178,120	12.19	OGBARU LGA				
11	Ikenga	24	8760	0.60	67	Akili Ogidi	320	116,800	7.99
12	Uga	1427	520,855	35.64	68	Akili Ozizor	1253	457,163	31.28
13	Umuchu	4441	1,620,965	110.90	69	Atani	3207	1,170,482	80.08
14	Ogboji	12	4380	0.30	70	Odekpe	2500	912,500	62.43
ANAMBRA EAST LGA					71	Ogbakuba	216	216	78,840
15	Aguleri	724	264,078	18.07	72	Okpoko	653	238,163	16.29
16	Igbariam	531	193,724	13.25	73	obeagwe	680	248,200	16.98
17	Nando	1151	420,206	28.75	74	Ossomala	28,312	1,033,388	707.01
18	Nsugbe	411	150,015	10.26	75	Umuodu	280	102,200	6.99
19	Umuleri	618	225,570	15.43	NNEWI NORTH LGA				
ANAOCHA LGA					76	Nnewichi	2618	955,585	66
20	Adazi-Ani	1764	643,676	44.04	77	Uruagu	542	197,680	14
21	Adazi-Enu	1065	388,871	26.61	78	Umudim	1578	575,970	39
22	Agulu	1143	417,268	28.55	79	Otolo	5028	1,835,344	125
23	Neni	120	43,800	3.00	EKWUSIGO LGA				
24	Nri	564	205,860	14.08	80	Ichi	150.65	54,987	4
25	Adazi-Nnukwu	36	13,140	0.90	81	Ihembosi	306	111,763	8
AWKA NORTH LGA					82	Oraifite	695	253,821	17
26	Achalla	76	27,740	1.90	83	Ozubulu	5235	1,910,935	131
27	Amagu	197	71,769	4.91	NJIKOKA LGA				
28	Amansea	739	269,643	18.45	84	Abagana	3696	1,349,040	92.30
29	Amanuke	583	212,630	14.55	85	Abba	36	13,140	0.90
30	Isuanocha	1680	613,200	41.95	86	Enugu-Agidi	4452	1,624,980	111.18
31	Mgbaukwu	1737	634,165	43.39	87	Enugu-Ukwu	2954	1,078,400	73.78
32	Ebenebe	1722	628,679	43.01	88	Nawfia	11,952	4,362,480	298.47
IDEMILI NORTH					89	Nimo	1932	705,180	48.25
33	Eziowelle	3496	1,276,040	87.30	ONITSHA NORTH LGA				
34	Obosi	3828	1,397,220	95.59	90	Onitsha	10,939	3,992,735	273.18
35	Orakwu	4298	1,568,916	107.34	ONITSHA SOUTH LGA				
36	Umuoji	3950	1,441,677	98.63	91	Fegge	7085.9	258,6371	176.95
37	Uke	1764	643,860	44.05	IHIALA				
38	Abatete	2214	808,110	55.29	92	Azia	859.45	313,699	21.46
39	Nkpor	4343	1,585,122	108.45	93	Ihiala	1789	652,985	44.68
40	Ogidi	1957	714,247	48.87	94	Isseke	60	21,900	1.50
IDEMILI SOUTH					95	Mbosi	170	62,050	98
41	Nnobi	744	271,378	18.57	96	Okija	268	97,820	6.69
42	Awka-Etiti	596	217,467	14.88	97	Orsumoghu	306	111,836	7.65
43	Oba	299	109,135	7.47	98	Uli	644	234,968	16.08
44	Akwu-Ukwu	493	179,854	12.30	AWKA SOUTH LGA				
45	Ojoto	221	80,537	5.51	99	Amawbia	2016	735,840	50.34
46	Nnokwa	831	303,315	20.75	100	Nibo	7628	2,784,220	190.49
47	Alor	229	83,403	5.71	101	Okpuno	620	226,300	15.48
DUNUKOFIA LGA					102	Awka	7814	2,852,110	195.13
48	Ukwulu	1236	451,140	30.88	103	Mgbaukwu	359.5	131,218	8.98
49	Ukpo	7246	2,644,790	181.06	104	Nise	728	265,720	18.18
50	Nawgu	3114	1,136,610	77.81	AYAMELUM LGA				
51	Ifitedunu	4132	1,508,180	103.25	105	Anaku	44	16,060	1.10
52	Umunachi	3202	1,168,730	80.01	106	Omor	517.9	189,034	12.93
NNEWI SOUTH LGA					107	Ifite-Ogwari	109.8	40,077	2.74
53	Osumuanyi	577	210,769	14.42	OYI LGA				
54	Ukpor	185	67,452	4.61	108	Umunya	1400	511,000	34.96
55	Amichi	1694	618,171	42.29	109	Awkuzu	327.68	119,603	8.18
ORUMBAH NORTH LGA					110	Nteje	6055	2,210,250	151.22
56	Ajalli	960	350,400	23.97	111	Nkwelle Ezunaka	120	43,800	3.00
57	Amaokpala	480	175,200	11.99	112	Ogbunike	120	43,800	3.00

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